

Analyzing Traffic Incidents in Izmir

Efe Korhan Uz
Industrial Engineering
Izmir University of Economics
Izmir, Turkey
efe.uz@std.ieu.edu.tr

Hüseyin Atacan Akgün
Software Engineering
Izmir University of Economics
Izmir, Turkey
huseyin.akgun@std.ieu.edu.tr

Hüseyin Yontar
Software Engineering
Izmir University of Economics
Izmir, Turkey
huseyin.yontar@std.ieu.edu.tr

Abstract—The abstract goes here.
Our code for this project is available at ¹.

I. INTRODUCTION

Traffic incidents are a prevalent problem in Izmir, imposing substantial costs on both citizens and the municipality, including life-threatening risks, economic losses, and operational burdens. Providing concise, data-driven evidence can accelerate decision-making and lead to precautionary actions that lower incident rates. While national studies exist, recent work centered on Izmir's incident patterns is scarce. This study analyzes a dataset of traffic incidents from the Izmir metropolitan area.

In this study, the incident records from the Izmir Municipality Open Data Portal are analyzed, including timestamps, location data and incident information. Our goal is to summarize where and when incident risk concentrates across the metropolitan area.

II. BACKGROUND AND RELATED WORK

Numerous studies have investigated the spatial and temporal relationships of traffic incident occurrence using various analytical and data-driven methods.

In a study conducted by Çakıcı on Izmir [1], it is shown that the highest number of accidents in the years between 2020-2023 occurred in the autumn and summer seasons. In addition, the least number of traffic accidents that occurred during that period was in the 2020 spring season due to the COVID-19 pandemic.

Another study conducted by Haybat and Karakaş [2] reveals that incidents are focused on the high mobility regions of Izmir, especially in the south of Karşıyaka district, all over district of Konak, northeast of Karabağlar district, and northwest of Buca. Furthermore, the study reveals that while the number of incidents increased between 2010 and 2014, the total coverage area declined, suggesting a higher spatial density of incidents in later years.

According to a study performed by Kumar and Toshniwal [3] on road accidents in Gujarat, the time periods between 20:00 to 22:00 and 22:00 to 00:00 were identified as the most accident-prone intervals when the 24-hour day was divided into two-hour periods.

III. DATA SET DESCRIPTION

The data set used in this study was provided by the Open Data Portal of Izmir Metropolitan Municipality. The data set includes detailed records of vehicle breakdowns and traffic accidents that occurred within the metropolitan area of Izmir. The observations cover the period between December 1, 2021, and September 28, 2025, and the data set was last updated on October 3, 2025. The raw data set consists of 21161 samples and 7 attributes. Each sample contains the date of the incident, the street name, the direction of travel, the location description, the type of incident, the accident time, and the intervention time. The descriptive information regarding the data set can be found in tabular form in Table I.

IV. PREPROCESSING

Describe the preprocessing steps that you have applied to your data set.

V. CLASSIFICATION / REGRESSION

Describe the classification / regression algorithms that you have applied to your data set. Present and discuss your results

VI. ASSOCIATION MINING

Describe the association mining algorithms that you have applied to your data set Present and discuss your results

VII. CLUSTERING

Describe the clustering algorithms that you have applied to your data set Present and discuss your results

VIII. ENSEMBLE LEARNING

Describe the ensemble learning algorithms that you have applied to your data set Present and discuss your results

IX. CONCLUSION

Conclude your report by summarising your findings

ACKNOWLEDGMENT

The authors would like to thank...

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TABLE I
DESCRIPTION OF EACH FEATURE IN THE RAW DATA SET.

Attribute	Type	Mean	Median	Mode	Number of Distinct Values	Number of Missing Values	Minimum	Maximum
Date	Interval	-	2023-07-18	2022-04-29	1397	0	2021-12-01	2025-09-28
Street	Nominal	-	-	Yeşildere Caddesi	183	0	-	-
Direction	Nominal	-	-	Merkez	157	16	-	-
Location	Nominal	-	-	Hilal Köprü Üstü	2032	14	-	-
Incident Type	Nominal	-	-	Maddi Hasarlı	30	0	-	-
Incident Time	Interval	-	15:30	17:29	1344	0	00:00	23:59
Intervention Time	Interval	-	15:52	18:10	1295	5193	00:00	23:59

REFERENCES

- [1] Z. Cakici, "Seasonal analysis of the distribution of urban traffic accidents: The case of İzmir province," in *5th International Conference on Innovative Academic Studies*. Konya, Türkiye: International Conference on Innovative Academic Studies, October 2024.
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- [3] S. Kumar and D. Toshniwal, "Analysis of hourly road accident counts using hierarchical clustering and cophenetic correlation coefficient (cpcc)," *Journal of Big Data*, vol. 3, no. 1, jul 2016. [Online]. Available: <https://journalofbigdata.springeropen.com/articles/10.1186/s40537-016-0046-3>